

Table 3: Instructions for installing BeautifulSoup

Usage	Instruction
Linux package manager	\$ sudo apt-get install python-bs4
PIP package manager of Python	\$ sudo pip install beautifulsoup4
Using <i>easy install</i> in Python	\$ sudo easy_install beautifulsoup4
Using <i>setup</i> file in source of Python package	\$ python setup.py install
To verify the successful installation	Python Shell> from bs4 import BeautifulSoup

The following code snippet can be used to extract images and content associated with the specific hashtag of Instagram. These can be beneficial for researchers working on specific news coverage linked with the particular hashtag. In addition, crime investigation authorities can extract all the messages or files linked with the specific hashtag from the timeline of users.

```
html = urllib.request.
urlopen(url, context=self.ctx).read()
soup = BeautifulSoup(html,
'html.parser')
myscript = soup.find('myscript',
text=lambda t: \

t.startswith('window._sharedData'))
page_json = myscript.text.
split(' = ', 1)[1].rstrip(';')
data = json.loads(page_json)
for instapost in data['entry_
data']['TagPage'][0]['graphql'
]['ihashtag']['edge_
ihashtag_to_media']['edges']:
    image_src =
instapost['node']['thumbnail_
resources'][1]['src']
    myhs = open(ihashtag +
'.txt', 'a')
    myhs.write(image_src +
'\n')

    myhs.close()
def main(self):
    self.ctx = ssl.create_default_
context()
    self.ctx.check_hostname = False
```

```
self.ctx.verify_mode = ssl.
CERT_NONE
with open('ihashtagdata.txt')
as f:
    self.content =
f.readlines()
    self.content = [x.strip() for x
in self.content]
    for ihashtag in self.content:
        self.getlinks(ihashtag,
'https://www.instagram.com/explore/
tags/'+ ihashtag + '/')
```

In addition, Twitter also provides a specific library called Tweepy for Web scraping, with the focus on R&D. Using this package, all the tweets can be extracted from social media about a particular search keyword mentioned in the code.

The following code snippet can extract all the tweets about specific search keywords from Twitter using the Tweepy package.

```
twitterconsumer_key =
*****
twitterconsumer_secret =
*****
twitteraccess_token =
*****
*****
twitteraccess_secret =
```

```
*****
class TweetListener(StreamListener):
    def on_currentdata(self,
currentdata):
        print currentdata
        return True
    def on_error(self, status):
        print status
oauth = OAuthHandler(twitterconsumer_
key, twitterconsumer_secret)
oauth.set_access_token(twitteraccess_
token, twitteraccess_secret)
mystream = Stream(auth,
TweetListener())
mystream.filter(track=['Search
Keywords'])
```

These types of scripts can be used to identify discussions around a particular keyword on social media. The police investigation teams and cyber forensic departments can analyse the tweets and messages which are related to a particular crime, so that deeper analysis of the persons involved can be done. These tweets are extracted in JavaScript Object Notation (JSON) format, which is further transformed to Comma Separated Value (CSV) or any other format, so that deep evaluation can be done.

Scope for R&D

Real-time Web data extraction is required for many applications including crime data analysis, market research, citizen feedback, consumer reviews, etc. Market research experts and practitioners can extract data about particular themes or products from multiple portals for user sentiment/behavioural analysis. Many corporate organisations collect such data sets from online portals so that they can improve their services and product features. **END** 

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